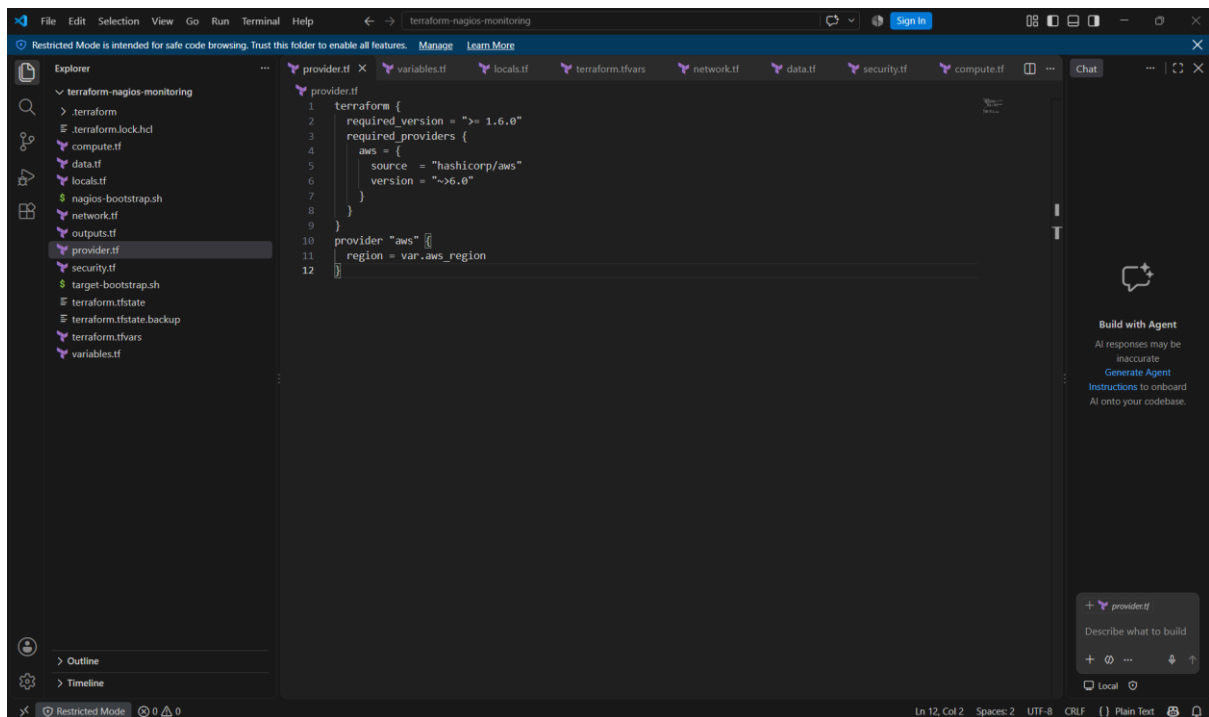
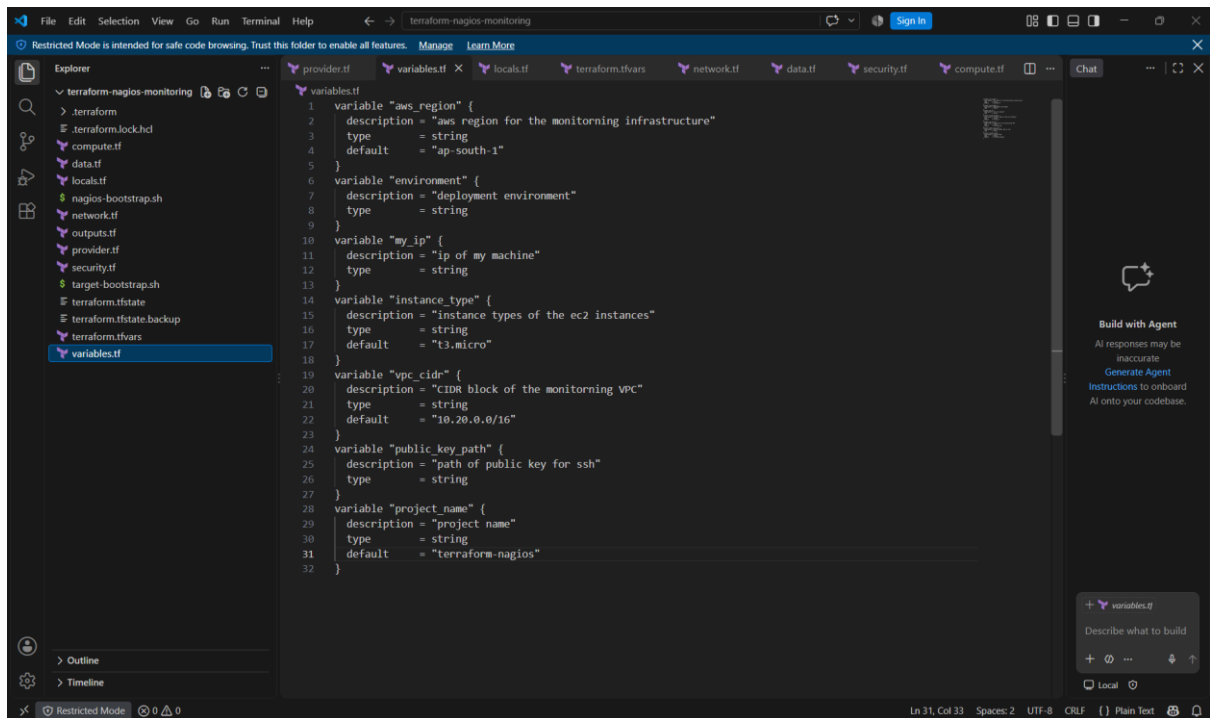
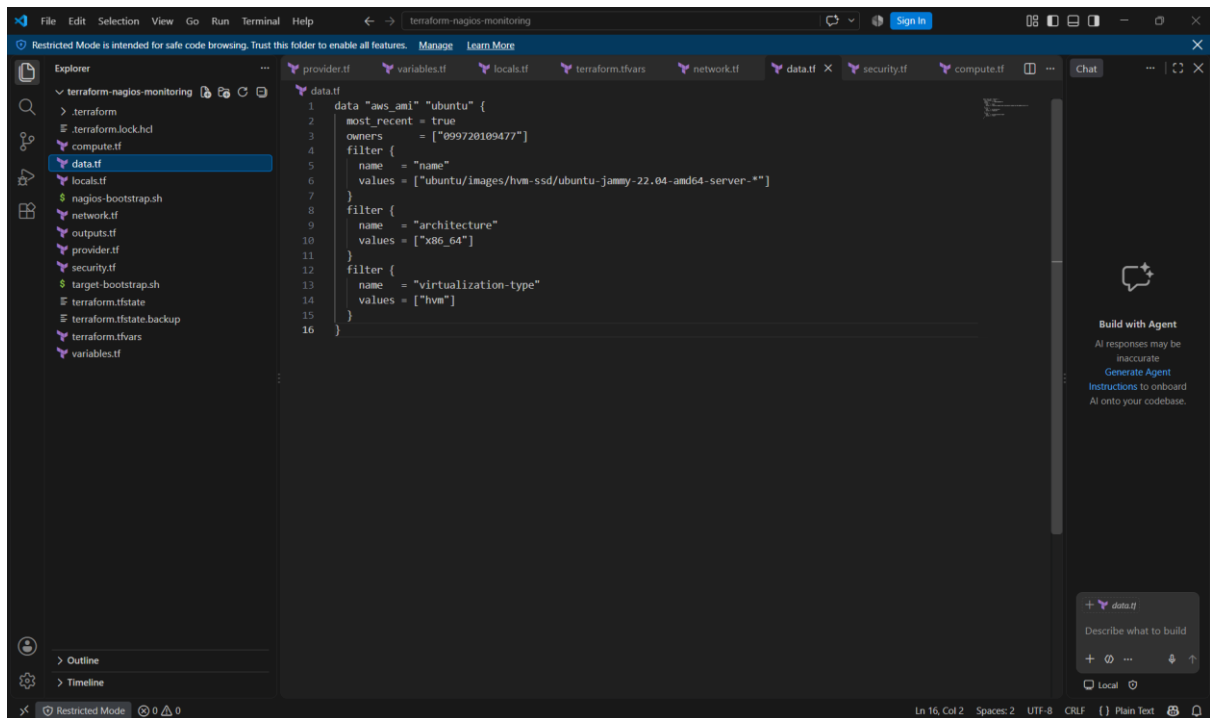
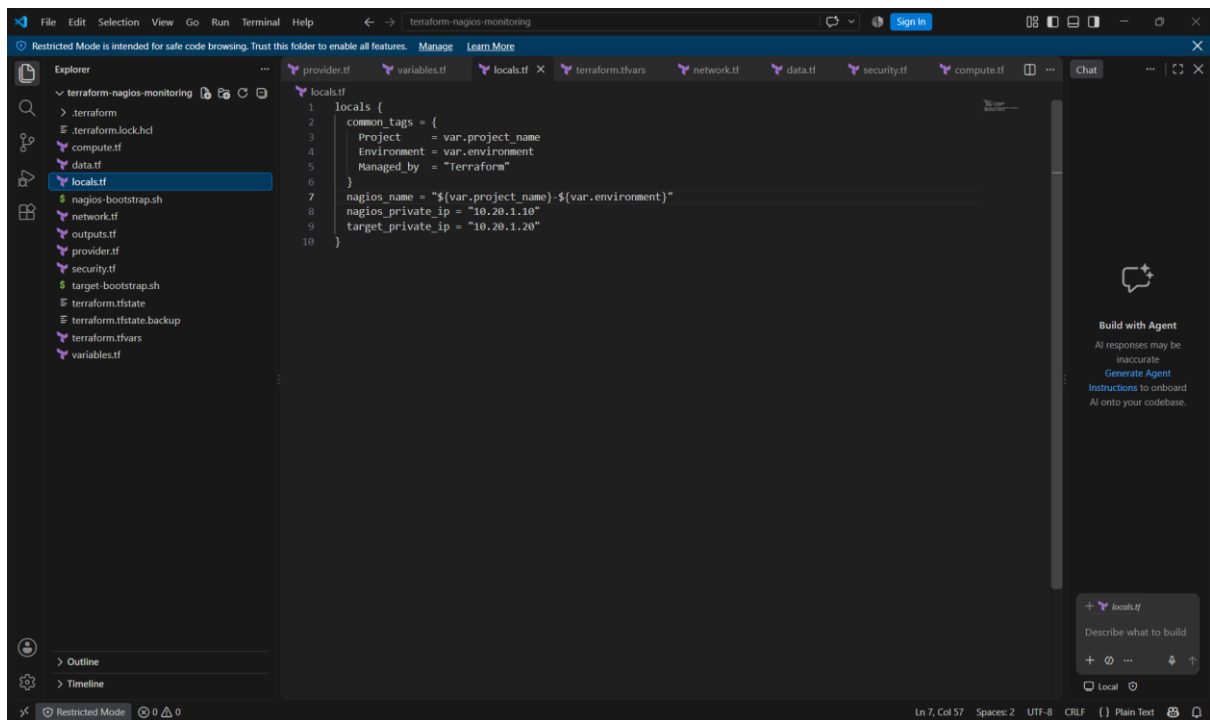
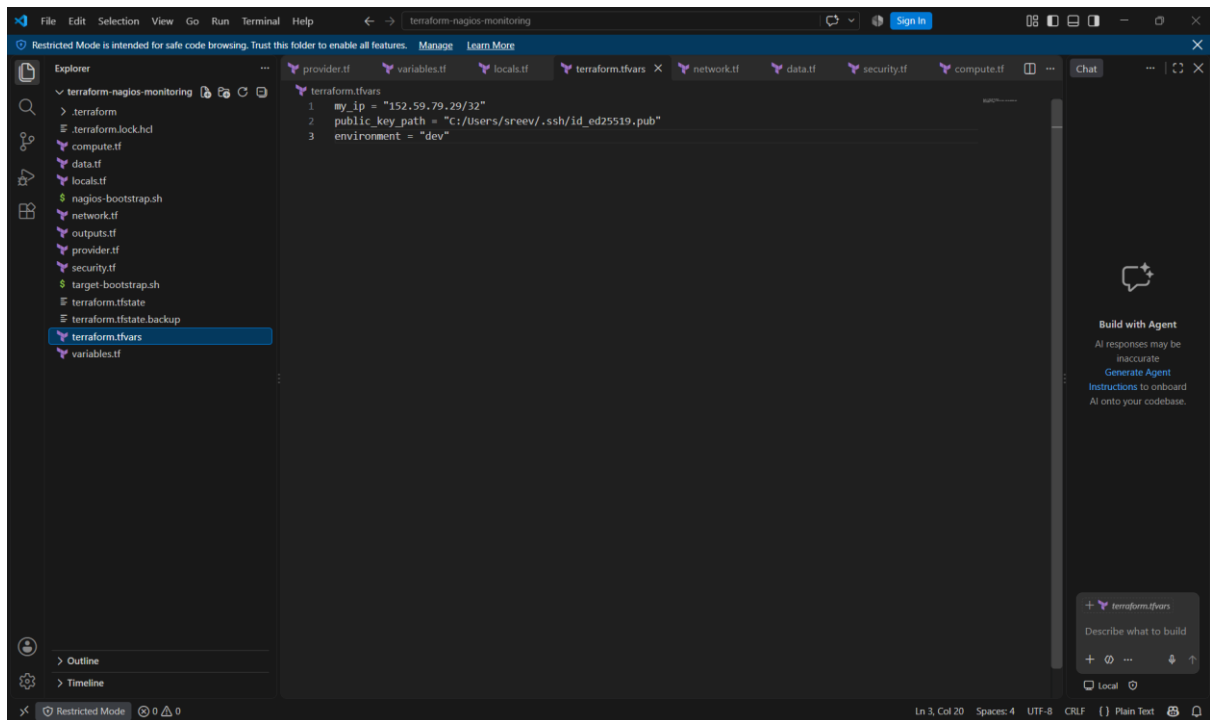


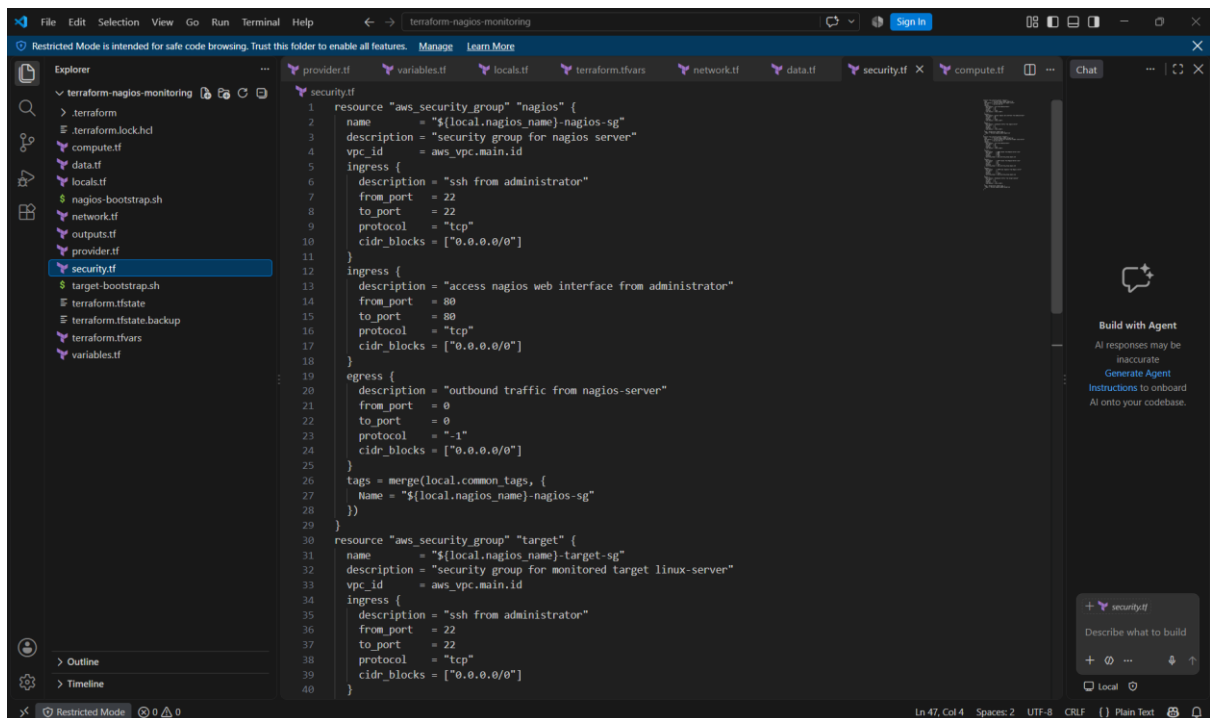
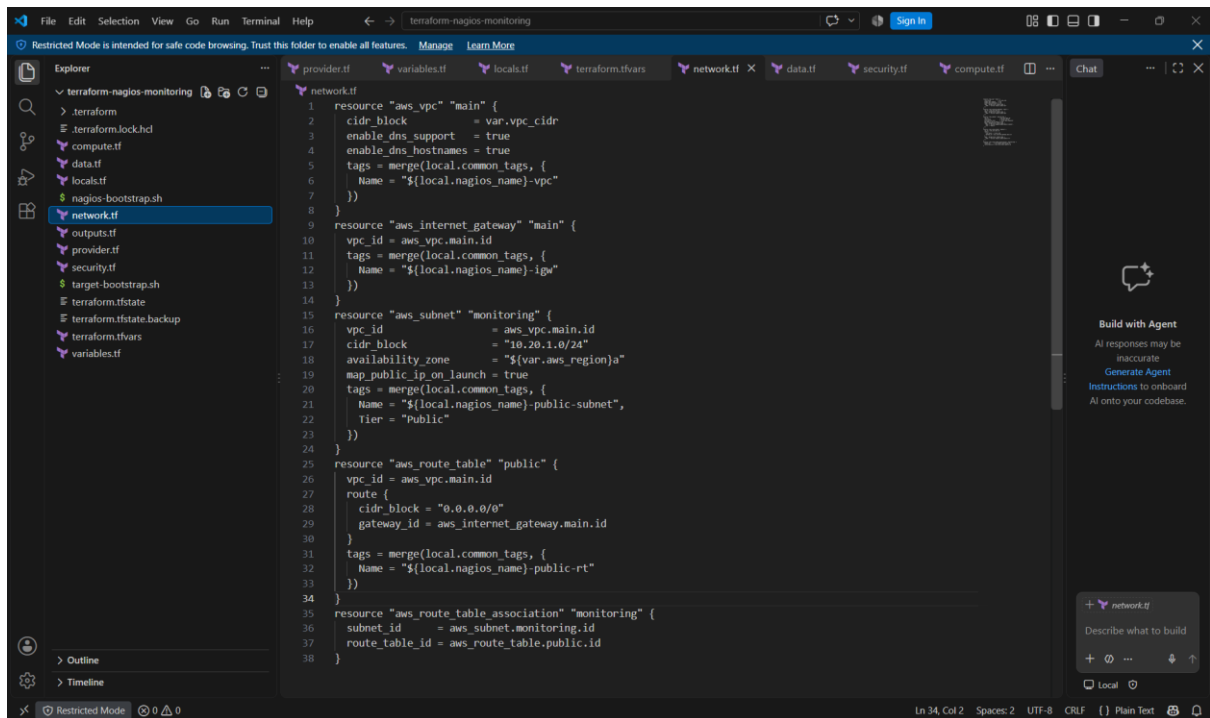
Flagship Practical-2 (Terraform+Nagios):

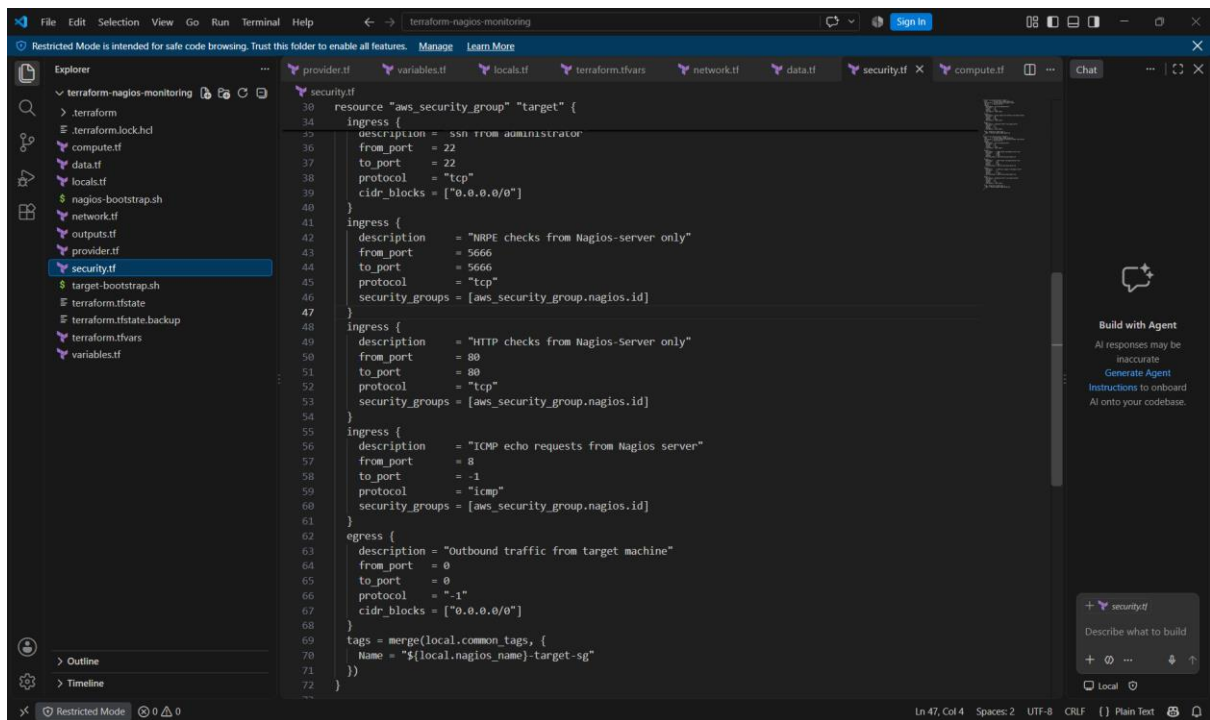
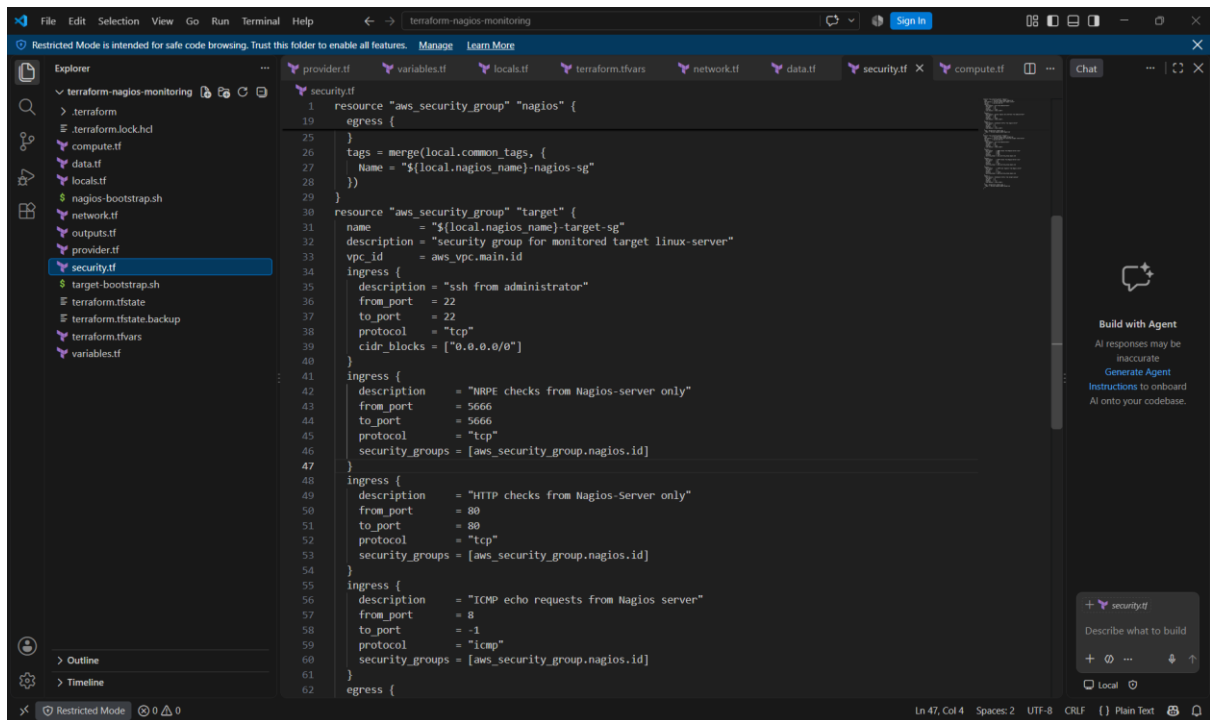
End-to-End AWS Infrastructure Automation and Enterprise-Style Nagios Monitoring with Terraform: VPC Architecture, Subnet Segmentation, Internet Gateway Routing, Secure EC2 Provisioning, SSH Access, Custom Security Groups, Automated Bootstrap Configuration, Nagios Core Deployment, NRPE-Based Remote Monitoring, Apache Web Dashboard, and Real-Time Host and Service Health Checks

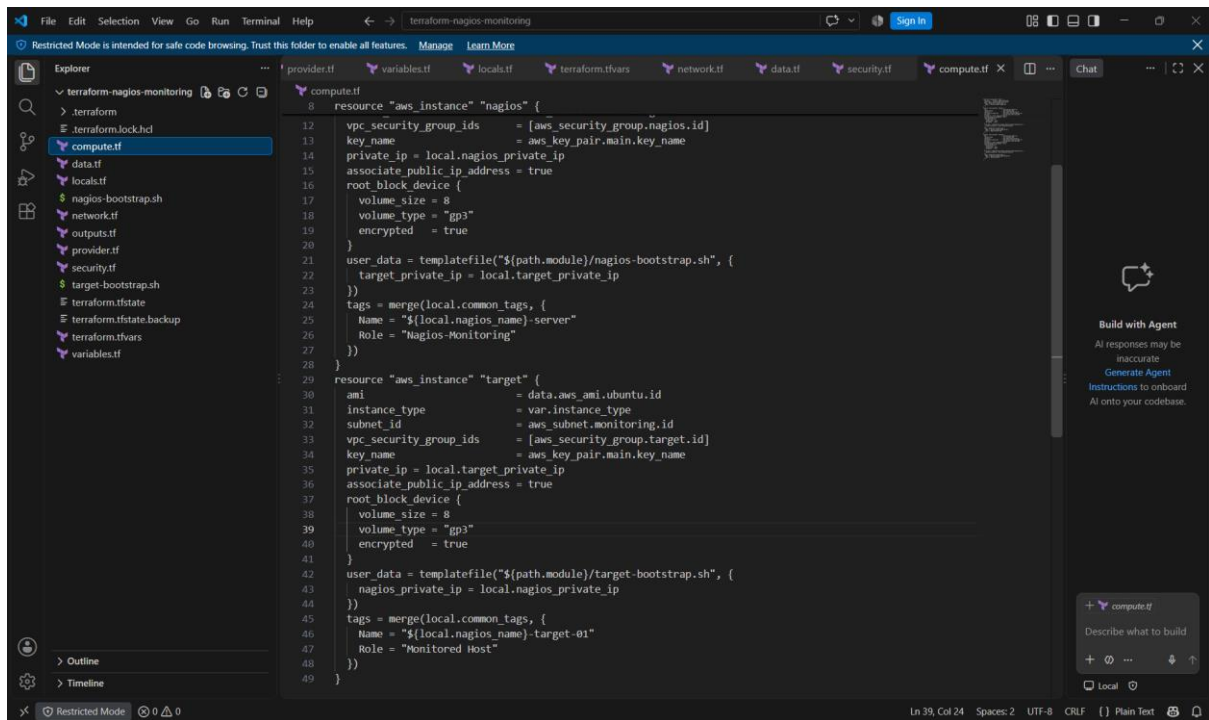
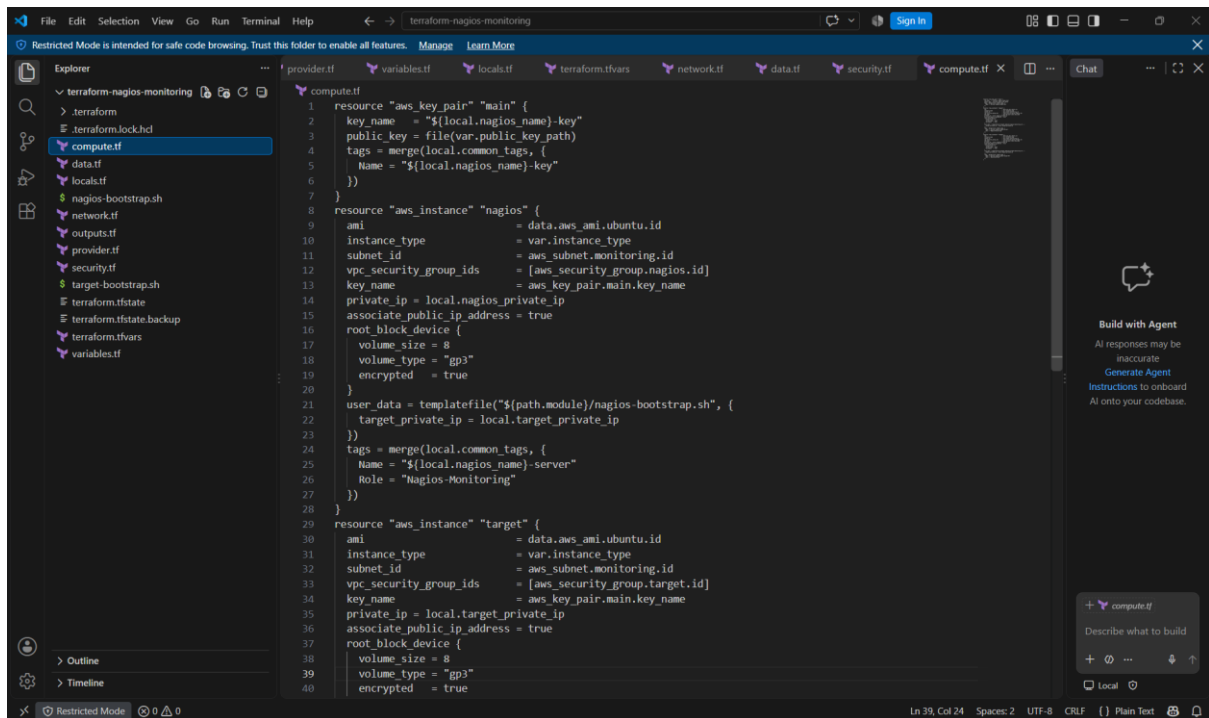












File Edit Selection View Go Run Terminal Help

terraform-nagios-monitoring

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08

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terraform.tfvars network.tf data.tf security.tf compute.tf outputs.tf nagios-bootstrap.sh Chat

terraform-nagios-monitoring .terraform .terraform.lock.hcl compute.tf data.tf locals.tf nagios-bootstrap.sh network.tf outputs.tf provider.tf security.tf target-bootstrap.sh terraform.tfstate terraform.tfstate.backup terraform.tfvars variables.tf

Outline Timeline

```
$ nagios-bootstrap.sh
1 #!/bin/bash
2 set -e; set -o pipefail
3 exec >> /dev/null; tee -a /var/log/nagios-bootstrap.log 2>&1
4 export DEBIAN_FRONTEND=noninteractive
5 apt-get update
6 apt-get install -y nagios4 nagios4-cgi nagios-nrpe-plugin monitoring-plugins apache2
7 azenmod cgi
8 azenconf nagios4-cgi || true
9 cat > /etc/nagios4/conf.d/target1.cfg <<'EOF'
10 define host {
11     use                generic-host
12     host_name           target1
13     alias               Terraform Nagios Target 01
14     address              ${target_private_ip}
15     check_command       check-host-alive
16     max_check_attempts  3
17     check_interval      1
18     retry_interval      1
19 }
20
21 define service {
22     use                generic-service
23     host_name           target1
24     service_description PING
25     check_command       check_ping!100.0,20%!500.0,60%
26 }
27
28 define service {
29     use                generic-service
30     host_name           target1
31     service_description HTTP
32     check_command       check_http
33 }
34
35 define service {
36     use                generic-service
37     host_name           target1
38     service_description NRPE load
39     check_command       check_nrpe!check_load
40 }
```

Build with Agent
AI responses may be inaccurate
Generate Agent
Instructions to onboard AI onto your codebase.

+ nagios-bootstrap.sh
Describe what to build
+ ...
Local

Ln 16, Col 30 Spaces: 4 UTF-8 LF Shell Script

File Edit Selection View Go Run Terminal Help

terraform-nagios-monitoring

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terraform.tfvars network.tf data.tf security.tf compute.tf outputs.tf nagios-bootstrap.sh Chat

terraform-nagios-monitoring .terraform .terraform.lock.hcl compute.tf data.tf locals.tf nagios-bootstrap.sh network.tf outputs.tf provider.tf security.tf target-bootstrap.sh terraform.tfstate terraform.tfstate.backup terraform.tfvars variables.tf

Outline Timeline

```
$ nagios-bootstrap.sh
10 define host {
16     max_check_attempts 3
17     check_interval      1
18     retry_interval      1
19 }
20
21 define service {
22     use                generic-service
23     host_name           target1
24     service_description PING
25     check_command       check_ping!100.0,20%!500.0,60%
26 }
27
28 define service {
29     use                generic-service
30     host_name           target1
31     service_description HTTP
32     check_command       check_http
33 }
34
35 define service {
36     use                generic-service
37     host_name           target1
38     service_description NRPE load
39     check_command       check_nrpe!check_load
40 }
41
42 define service {
43     use                generic-service
44     host_name           target1
45     service_description NRPE Disk
46     check_command       check_nrpe!check_disk
47 }
48 EOF
49 nagios4 -v /etc/nagios4/nagios.cfg
50 systemctl enable --now nagios4
51 systemctl enable --now apache2
52 systemctl restart nagios4
53 systemctl restart apache2
54
```

Build with Agent
AI responses may be inaccurate
Generate Agent
Instructions to onboard AI onto your codebase.

+ nagios-bootstrap.sh
Describe what to build
+ ...
Local

Ln 16, Col 30 Spaces: 4 UTF-8 LF Shell Script

The screenshot shows a code editor with a file explorer on the left and a chat sidebar on the right. The file explorer shows a directory structure for 'terraform-nagios-monitoring' with files like 'network.tf', 'data.tf', 'security.tf', 'compute.tf', 'outputs.tf', 'nagios-bootstrap.sh', 'target-bootstrap.sh', 'provider.tf', 'security.tf', 'variables.tf', 'terraform.tfstate', 'terraform.lock.hcl', 'terraform.tfvars', and 'variables.tf'. The 'target-bootstrap.sh' file is selected and its content is displayed in the main editor. The script is a shell script that sets up Nagios on a target machine. It includes commands for setting the shell to bash, setting the pipefail option, setting the log file, setting the DEBIAN_FRONTEND to noninteractive, updating apt, installing apache2, nagios-nrpe-server, and monitoring-plugins, enabling the services, setting the allowed hosts in the nagios.cfg file, setting the dont_blame_nrpe option, restarting the nagios-nrpe-server, creating an index.html file, and restarting apache2.

```
1 #!/bin/bash
2 set -euo pipefail
3 exec >> (tee -a /var/log/target-bootstrap.log) 2>&1
4 export DEBIAN_FRONTEND=noninteractive
5 apt-get update
6 apt-get install -y apache2 nagios-nrpe-server monitoring-plugins
7 systemctl enable --now nagios-nrpe-server
8 systemctl enable --now apache2
9 sed -i 's/allowed_hosts=.*allowed_hosts=127.0.0.1,${nagios_private_ip}/' /etc/nagios/nrpe.cfg
10 sed -i 's/dont_blame_nrpe=.*dont_blame_nrpe=0/' /etc/nagios/nrpe.cfg
11 systemctl restart nagios-nrpe-server
12 cat > /var/www/html/index.html <<'EOF'
13 <html>
14 <body>
15 <h1>Target server is operational</h1>
16 </body>
17 </html>
18 EOF
19 systemctl restart apache2
```

The screenshot shows the same code editor with the 'outputs.tf' file selected. The file contains Terraform output definitions for various Nagios components. The outputs include the VPC ID, Nagios instance ID, Nagios public IP, Nagios private IP, target instance ID, target public IP, target private IP, and the Nagios dashboard URL. The dashboard URL is constructed using the Nagios public IP and the Nagios instance ID.

```
1 output "vpc_id" {
2   description = "ID of the Nagios monitoring VPC"
3   value       = aws_vpc.main.id
4 }
5
6 output "nagios_instance_id" {
7   description = "EC2 instance ID of the Nagios server"
8   value       = aws_instance.nagios.id
9 }
10
11 output "nagios_public_ip" {
12   description = "Public IPv4 address of the Nagios server"
13   value       = aws_instance.nagios.public_ip
14 }
15
16 output "nagios_private_ip" {
17   description = "Private IPv4 address of the Nagios server"
18   value       = aws_instance.nagios.private_ip
19 }
20
21 output "target_instance_id" {
22   description = "EC2 instance ID of the monitored host"
23   value       = aws_instance.target.id
24 }
25
26 output "target_public_ip" {
27   description = "Public IPv4 address of the monitored host"
28   value       = aws_instance.target.public_ip
29 }
30
31 output "target_private_ip" {
32   description = "Private IPv4 address of the monitored host"
33   value       = aws_instance.target.private_ip
34 }
35
36 output "nagios_dashboard_url" {
37   description = "Nagios dashboard URL"
38   value       = "http://${aws_instance.nagios.public_ip}/nagios"
39 }
```

```
Windows PowerShell
aws_instance.nagios: Creating...
aws_instance.target: Destruction complete after 11s
aws_instance.target: Creating...
aws_instance.nagios: Still creating... [00m10s elapsed]
aws_instance.target: Still creating... [00m10s elapsed]
aws_instance.target: Creation complete after 15s [id=i-08a3ff32a79b70564]
aws_instance.nagios: Creation complete after 15s [id=i-0155d325754cc4b1a]

Apply complete! Resources: 2 added, 0 changed, 2 destroyed.

Outputs:

nagios_dashboard_url = "http://13.233.216.192/nagios"
nagios_instance_id = "i-0155d325754cc4b1a"
nagios_private_ip = "10.20.1.10"
nagios_public_ip = "13.233.216.192"
target_instance_id = "i-08a3ff32a79b70564"
target_private_ip = "10.20.1.20"
target_public_ip = "15.207.87.77"
vpc_id = "vpc-0e07430cb1c036a44"
PS D:\terraform-nagios-monitoring> terraform output
nagios_dashboard_url = "http://13.233.216.192/nagios"
nagios_instance_id = "i-0155d325754cc4b1a"
nagios_private_ip = "10.20.1.10"
nagios_public_ip = "13.233.216.192"
target_instance_id = "i-08a3ff32a79b70564"
target_private_ip = "10.20.1.20"
target_public_ip = "15.207.87.77"
vpc_id = "vpc-0e07430cb1c036a44"
PS D:\terraform-nagios-monitoring> |
```

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#VpcDetails:VpcId=vpc-0e07430cb1c036a44

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VPC > Your VPCs > vpc-0e07430cb1c036a44

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

- Network ACLs
- Security groups

PrivateLink and Lattice

- Getting started

vpc-0e07430cb1c036a44 / terraform-nagios-dev-vpc

Details Info

VPC ID vpc-0e07430cb1c036a44	State Available	Block Public Access Off	DNS hostnames Enabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-0ed98e351644b4b55	Main route table rtb-062168ca1701e8248
Main network ACL acl-0a7954ee760b99895	Default VPC No	IPv4 CIDR 10.20.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups Failed to load rule groups	Owner ID 368999587946
Encryption control ID -	Encryption control mode -		

Resource map Info

Subnets (1)
Subnets within this VPC

ap-south-1a

terraform-nagios-dev-public-subnet

Route tables (2)
Route network traffic to resources

rtb-062168ca1701e8248

terraform-nagios-dev-public-rt

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← → ↺ ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#VpcDetails:VpcId=vpc-0e07430cb1c036a44

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VPC > Your VPCs > vpc-0e07430cb1c036a44

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Virtual private cloud Your VPCs Subnets Route tables Internet gateways Egress-only internet gateways DHCP option sets Elastic IPs Managed prefix lists NAT gateways Peering connections Route servers

Security Network ACLs Security groups

PrivateLink and Lattice Getting started

VPC ID vpc-0e07430cb1c036a44

State Available

DNS resolution Enabled

Main network ACL acl-0a7954ee760b99895

IPv6 CIDR (Network border group) -

Encryption control ID -

Tenancy default

Default VPC No

Network Address Usage metrics Disabled

Encryption control mode -

Block Public Access Off

DHCP option set dopt-0ed98e351644b4b55

IPv4 CIDR 10.20.0.0/16

Route 53 Resolver DNS Firewall rule groups Failed to load rule groups

DNS hostnames Enabled

Main route table rtb-062168ca1701e8248

IPv6 pool -

Owner ID 368999587946

Resource map CIDRs Flow logs Tags Related resources Integrations

Resource map info Show all details

Subnets (1) Subnets within this VPC

ap-south-1a

terraform-nagios-dev-public-subnet

Route tables (2) Route network traffic to resources

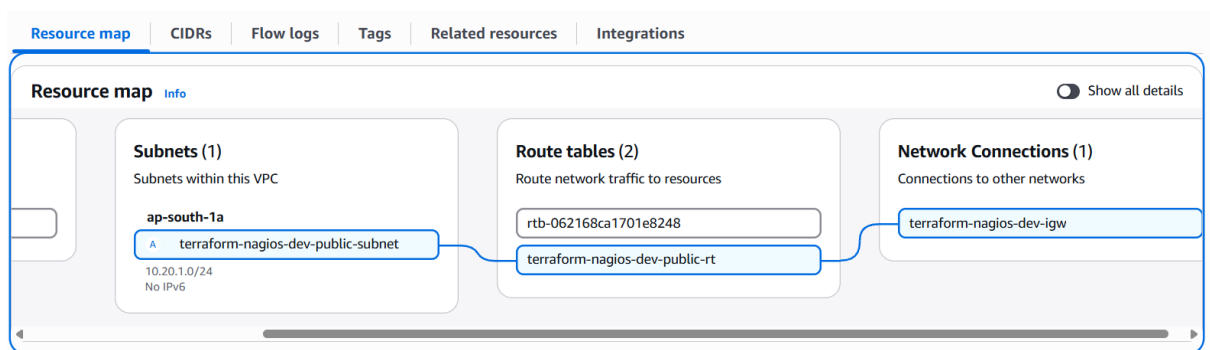
rtb-062168ca1701e8248

terraform-nagios-dev-public-rt

Network Connections (1) Connections to other networks

terraform-nagios-dev-igw

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Resource map	CIDRs	Flow logs	Tags	Related resources	Integrations
IPv4 CIDRs Info Edit CIDRs					
Address family	CIDR	Status			
IPv4	10.20.0.0/16	Associated			

Resource map	CIDRs	Flow logs	Tags	Related resources	Integrations
Tags Manage tags					
Search tags					
< 1 >					
Key	Value				
Managed_by	Terraform				
Name	terraform-nagios-dev-vpc				
Environment	dev				
Project	terraform-nagios				

← → ↺

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SubnetDetails:subnetId=subnet-0ce2c7ed754b3cf7f

☆ + 🇮🇳 🇮🇳 🇮🇳 🇮🇳 🇮🇳 🇮🇳 🇮🇳 🇮🇳 🇮🇳 🇮🇳

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Account ID: 3689-9958-7946

tf-admin

VPC > Subnets > subnet-0ce2c7ed754b3cf7f

VPC dashboard <

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Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

subnet-0ce2c7ed754b3cf7f / terraform-nagios-dev-public-subnet [Actions](#)

Details

Subnet ID

Subnet ARN

State

Block Public Access

IPv4 CIDR

Available IPv4 addresses

IPv6 CIDR

VPC

Auto-assign public IPv4 address

Outpost ID

Hostname type

Owner

Availability Zone

Network ACL

Auto-assign customer-owned IPv4 address

IPv6 CIDR reservations

Resource name DNS AAAA record

Default subnet

Customer-owned IPv4 pool

IPv6-only

DNS64

IPv6 CIDR association ID

Route table

Auto-assign IPv6 address

IPv4 CIDR reservations

Resource name DNS A record

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Related resources

Route table: rtb-0fc4d90550597e929 / terraform-nagios-dev-public-rt [Edit route table association](#)

Routes (2)

Filter routes

< 1 >

Destination

Target

10.20.0.0/16

local

0.0.0.0/0

[igw-0ed25802037193071](#)

CloudShell

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Flow logs	Route table	Network ACL	CIDR reservations	Sharing	Tags	Related resources
Route table: rtb-0fc4d90550597e929 / terraform-nagios-dev-public-rt Edit route table association						
Routes (2)						
Filter routes						
< 1 >						
Destination	Target					
10.20.0.0/16	local					
0.0.0.0/0	igw-0ed25802037193071					

← → ↻ ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#InternetGateway:internetGatewayId=igw-0ed25802037193071 ☆ 🌐 📄 🗑️ 🔍

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VPC > Internet gateways > igw-0ed25802037193071 ⓘ 🖨️

VPC dashboard <

Filter by VPC ▾

[AWS Global View](#)

▼ **Virtual private cloud**

- Your VPCs
- Subnets
- Route tables
- Internet gateways**
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

▼ **Security**

- Network ACLs
- Security groups

▼ **PrivateLink and Lattice**

- Getting started

igw-0ed25802037193071 / terraform-nagios-dev-igw Actions ▾

Details [Info](#)

Internet gateway ID
igw-0ed25802037193071

State
Attached

VPC ID
vpc-0e07430cb1c036a44 | [terraform-nagios-dev-vpc](#)

Owner
368999587946

Tags (4) Manage tags

Search tags

Key	Value
Environment	dev
Project	terraform-nagios
Managed_by	Terraform
Name	terraform-nagios-dev-igw

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← → ↻ ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#InstanceDetails:instanceId=i-0155d325754cc4b1a ☆ 🌐 📄 🗑️ 🔍

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EC2 > Instances > i-0155d325754cc4b1a ⓘ 🖨️

EC2 <

- Dashboard
- [AWS Global View](#)
- Events
- ▼ **Instances**
- Instances
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- UltraServers [New](#)
- Capacity Manager [New](#)
- ▼ **Images**
- AMIs
- AMI Catalog
- ▼ **Elastic Block Store**
- Volumes
- Snapshots
- Lifecycle Manager

Instance summary for i-0155d325754cc4b1a (terraform-nagios-dev-server) Info Connect Instance state ▾ Actions ▾

Updated less than a minute ago

Instance ID
i-0155d325754cc4b1a

IPv6 address
-

Hostname type
IP name: ip-10-20-1-10.ap-south-1.compute.internal

Answer private resource DNS name
-

Auto-assigned IP address
13.233.216.192 [Public IP]

IAM role
-

IMDSv2
Optional
⚠️ [EC2 recommends setting IMDSv2 to required](#)

Public IPv4 address
13.233.216.192 | [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-10-20-1-10.ap-south-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-0e07430cb1c036a44 (terraform-nagios-dev-vpc)

Subnet ID
subnet-0ce2c7ed754b3cf7f (terraform-nagios-dev-public-subnet)

Instance ARN
arn:aws:ec2:ap-south-1:368999587946:instance/i-0155d325754cc4b1a

Private IPv4 addresses
10.20.1.10

Public DNS
ec2-13-233-216-192.ap-south-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses
-

AWS Compute Optimizer finding
⚠️ User: arn:aws:iam:368999587946:user/tf-admin is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action
[Retry](#)

Auto Scaling Group name
-

Managed
false

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The screenshot shows the AWS Management Console for a security group named `sg-096c729e38d8faddf` (terraform-nagios-dev-target-sg). The console displays the following rules:

Inbound rules						
Name	Security group rule ID	Port range	Protocol	Source	Security groups	
-	sg-0e19974f662f17f97	8 - -1	ICMP	sg-02c77b60df0dda629	terraform-nagios	
-	sg-0b9adb69d3d76c5ea	80	TCP	sg-02c77b60df0dda629	terraform-nagios	
-	sg-057e4b5c4cecc4764	22	TCP	0.0.0.0/0	terraform-nagios	
-	sg-002e479ee5a5b218d	5666	TCP	sg-02c77b60df0dda629	terraform-nagios	

Outbound rules						
Name	Security group rule ID	Port range	Protocol	Destination	Security groups	
-	sg-0112beef03b5dc97	All	All	0.0.0.0/0	terraform-nagios	

Note: Inbound and outbound rules for Target machine (port 5666 for NRPE and ICMP for ping requests to verify target up/down status by Nagios-Monitoring-server, source for ICMP, TCP 80 and TCP 5666 is nothing but the security group of Nagios-Monitoring-Server)

The screenshot shows the details of an EBS volume named `vol-0607bda6e09ec5956`. The details are as follows:

Details			
Volume ID vol-0607bda6e09ec5956	Size 8 GiB	Type gp3	Status check ✓ Okay
Owner ID 36899587946 (This account)	Volume state ✓ In-use	IOPS 3000	Throughput 125
Fast snapshot restored No	Availability Zone aps1-az1 (ap-south-1a)	Created Tue Sep 29 2026 20:14:17 GMT+0530 (India Standard Time)	Multi-Attach enabled No
Attached resources i-0155d325754cc4b1a (terraform-nagios-dev-server) (EBS card index: 0) /dev/sda1 (attached)	Outposts ARN -	Managed false	Operator -

AWS Compute Optimizer finding
 ⚠ This user is not authorized to call AWS Compute Optimizer. | [Retry](#)

Source	
Snapshot ID snap-0cb9e2e82472874e7	Source volume ID -

Encryption			
Encryption Encrypted	KMS key ID 8-c79259e-df0f-4492-9aeb-	KMS key alias -	KMS key ARN arn:aws:kms:ap-south-

AWS Management Console screenshot showing the details of an Amazon EBS Volume (vol-0d1a27344a23fa5c4).

Volume ID: vol-0d1a27344a23fa5c4

Size: 8 GiB

Type: gp3

Status check: ✔ Okay

Owner ID: 36899587946 (This account)

Volume state: ✔ In-use

IOPS: 3000

Throughput: 125

Fast snapshot restored: No

Availability Zone: ap-s1-az1 (ap-south-1a)

Created: Tue Sep 29 2026 20:14:17 GMT+0530 (India Standard Time)

Multi-Attach enabled: No

Attached resources: i-08a3ff52a79b70564 (terraform-nagios-dev-target-01) (EBS card index: 0); /dev/sda1 (attached)

Outposts ARN: -

Managed: false

Operator: -

AWS Compute Optimizer finding: ⚠ This user is not authorized to call AWS Compute Optimizer. | [Retry](#)

Source:

Snapshot ID: snap-0cb9e2e82472874e7

Source volume ID: -

Encryption: Encrypted

KMS key ID: 8c79259e-dc9f-4492-9eab-

KMS key alias: -

KMS key ARN: arnaws:kms:ap-south-

```
ubuntu@ip-10-20-1-10: ~
sreev@admin MINGW64 ~/.ssh (main)
$ ssh -i "id_ed25519" ubuntu@ec2-13-233-216-192.ap-south-1.compute.amazonaws.com
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1064-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Tue Sep 29 15:43:09 UTC 2026

System load:  0.0           Processes:            136
Usage of /:   28.6% of 7.57GB Users logged in:         1
Memory usage: 28%          IPv4 address for ens5: 10.20.1.10
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

3 updates can be applied immediately.
3 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm
```

```
ubuntu@ip-10-20-1-10: ~  
ubuntu@ip-10-20-1-10:~$ sudo tail -n 40 /var/log/nagios-bootstrap.log  
Reading configuration data...  
  Read main config file okay...  
  Read object config files okay...  
  
Running pre-flight check on configuration data...  
  
Checking objects...  
  Checked 12 services.  
Warning: Host 'target1' has no default contacts or contactgroups defined!  
  Checked 2 hosts.  
  Checked 1 host groups.  
  Checked 0 service groups.  
  Checked 1 contacts.  
  Checked 1 contact groups.  
  Checked 180 commands.  
  Checked 5 time periods.  
  Checked 0 host escalations.  
  Checked 0 service escalations.  
Checking for circular paths...  
  checked 2 hosts  
  Checked 0 service dependencies  
  Checked 0 host dependencies  
  Checked 5 timeperiods
```

```
ubuntu@ip-10-20-1-10: ~  
  Checked 0 service dependencies  
  Checked 0 host dependencies  
  Checked 5 timeperiods  
Checking global event handlers...  
Checking obsessive compulsive processor commands...  
Checking misc settings...  
  
Total Warnings: 1  
Total Errors: 0  
  
Things look okay - No serious problems were detected during the pre-flight check  
+ systemctl enable --now nagios4  
Synchronizing state of nagios4.service with SysV service script with /lib/systemd/  
d/systemd-sysv-install.  
Executing: /lib/systemd/systemd-sysv-install enable nagios4  
Created symlink /etc/systemd/system/multi-user.target.wants/nagios4.service → /l  
ib/systemd/system/nagios4.service.  
+ systemctl enable --now apache2  
Synchronizing state of apache2.service with SysV service script with /lib/system  
d/systemd-sysv-install.  
Executing: /lib/systemd/systemd-sysv-install enable apache2  
+ systemctl restart nagios4  
+ systemctl restart apache2  
ubuntu@ip-10-20-1-10:~$
```

```
ubuntu@ip-10-20-1-10: ~  
ubuntu@ip-10-20-1-10:~$ sudo cat /etc/nagios4/conf.d/target1.cfg  
define host {  
    use                generic-host  
    host_name          target1  
    alias              Terraform Nagios Target 01  
    address            10.20.1.20  
    check_command       check-host-alive  
    max_check_attempts 3  
    check_interval     1  
    retry_interval     1  
}  
  
define service {  
    use                generic-service  
    host_name          target1  
    service_description PING  
    check_command       check_ping!100.0,20%!500.0,60%  
}  
  
define service {  
    use                generic-service  
    host_name          target1  
    service_description HTTP  
    check_command       check_http
```

```
ubuntu@ip-10-20-1-10: ~  
    check_command       check_ping!100.0,20%!500.0,60%  
}  
  
define service {  
    use                generic-service  
    host_name          target1  
    service_description HTTP  
    check_command       check_http  
}  
  
define service {  
    use                generic-service  
    host_name          target1  
    service_description NRPE Load  
    check_command       check_nrpe!check_load  
}  
  
define service {  
    use                generic-service  
    host_name          target1  
    service_description NRPE Disk  
    check_command       check_nrpe!check_disk  
}  
ubuntu@ip-10-20-1-10:~$
```

```
ubuntu@ip-10-20-1-20: ~
sreev@admin MINGW64 ~/.ssh (main)
$ ssh -i "id_ed25519" ubuntu@ec2-15-207-87-77.ap-south-1.compute.amazonaws.com
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1064-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Sep 29 15:47:31 UTC 2026

System load:  0.0              Processes:            112
Usage of /:   27.7% of 7.57GB  Users logged in:     1
Memory usage: 26%             IPv4 address for ens5: 10.20.1.20
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

3 updates can be applied immediately.
3 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm
```

```
ubuntu@ip-10-20-1-20: ~
ubuntu@ip-10-20-1-20:~$ sudo systemctl status nagios-nrpe-server
● nagios-nrpe-server.service - Nagios Remote Plugin Executor
   Loaded: loaded (/lib/systemd/system/nagios-nrpe-server.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-09-29 15:21:44 UTC; 26min ago
     Docs: http://www.nagios.org/documentation
   Main PID: 6380 (nrpe)
    Tasks: 1 (limit: 1009)
   Memory: 1.5M
      CPU: 106ms
   CGroup: /system.slice/nagios-nrpe-server.service
           └─6380 /usr/sbin/nrpe -c /etc/nagios/nrpe.cfg -f

Sep 29 15:21:44 ip-10-20-1-20 systemd[1]: Started Nagios Remote Plugin Executor.
Sep 29 15:21:44 ip-10-20-1-20 nrpe[6380]: Starting up daemon
Sep 29 15:21:44 ip-10-20-1-20 nrpe[6380]: Server listening on 0.0.0.0 port 5666.
Sep 29 15:21:44 ip-10-20-1-20 nrpe[6380]: Server listening on :: port 5666.
Sep 29 15:21:44 ip-10-20-1-20 nrpe[6380]: Listening for connections on port 5666
Sep 29 15:21:44 ip-10-20-1-20 nrpe[6380]: Allowing connections from: 127.0.0.1,10.20.1.20
ubuntu@ip-10-20-1-20:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-09-29 14:45:28 UTC; 1h 3min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 6108 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
```

```
ubuntu@ip-10-20-1-20: ~  
ubuntu@ip-10-20-1-20:~$ sudo tail -n 40 /var/log/target-bootstrap.log  
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /lib/s  
ystemd/system/apache2.service.  
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.serv  
ice → /lib/systemd/system/apache-htcacheclean.service.  
Setting up samba-libs:amd64 (2:4.15.13+dfsg-0ubuntu1.13) ...  
Setting up libsmbclient:amd64 (2:4.15.13+dfsg-0ubuntu1.13) ...  
Setting up smbclient (2:4.15.13+dfsg-0ubuntu1.13) ...  
Setting up samba-dsdb-modules:amd64 (2:4.15.13+dfsg-0ubuntu1.13) ...  
Setting up python3-samba (2:4.15.13+dfsg-0ubuntu1.13) ...  
Setting up samba-common-bin (2:4.15.13+dfsg-0ubuntu1.13) ...  
Checking smb.conf with testparm  
Load smb config files from /etc/samba/smb.conf  
Loaded services file OK.  
Weak crypto is allowed  
  
Server role: ROLE_STANDALONE  
  
Done  
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...  
Processing triggers for man-db (2.10.2-1) ...  
Processing triggers for libc-bin (2.35-0ubuntu3.15) ...  
  
Running kernel seems to be up-to-date.
```

```
ubuntu@ip-10-20-1-20: ~  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
+ systemctl enable --now nagios-nrpe-server  
Synchronizing state of nagios-nrpe-server.service with SysV service script with /lib  
/systemd/systemd-sysv-install.  
Executing: /lib/systemd/systemd-sysv-install enable nagios-nrpe-server  
+ systemctl enable --now apache2  
Synchronizing state of apache2.service with SysV service script with /lib/systemd/sy  
stemd-sysv-install.  
Executing: /lib/systemd/systemd-sysv-install enable apache2  
+ sed -i 's/^allowed_hosts=.* /allowed_hosts=127.0.0.1,10.20.1.10/' /etc/nagios/nrpe.  
cfg  
+ sed -i 's/^dont_blame_nrpe=.* /dont_blame_nrpe=0/' /etc/nagios/nrpe.cfg  
+ systemctl restart nagios-nrpe-server  
+ cat  
+ systemctl restart apache2  
ubuntu@ip-10-20-1-20:~$ sudo ss -tlnp | grep 5666  
LISTEN 0      5            0.0.0.0:5666      0.0.0.0:*        users:(("nrpe",pid=6380,fd=4))  
LISTEN 0      5            [::]:5666        [::]:*          users:(("nrpe",pid=6380,fd=5))  
ubuntu@ip-10-20-1-20:~$
```

Page Tour

Page Tour

Page Tour

13.233.216.192/nagios4/

Machine LearningNeural Networks ar...Density-based clust...Fisher LDA-StanfordCompetitive Progra...CVB in Unsupervise...Red Hat AcademySystem Design Primer

All Bookmarks

Nagios®

General

HomeDocumentation

Current Status

Tactical OverviewMap (Legacy)HostsServicesHost GroupsSummaryGridService GroupsSummaryGridProblemsServices (Unhandled)Hosts (Unhandled)Network OutagesQuick Search:

Reports

AvailabilityTrends (Legacy)AlertsHistorySummaryHistogram (Legacy)NotificationsEvent Log

System

CommentsDowntimeProcess InfoPerformance InfoScheduling QueueConfiguration

Service Information

Last Updated: Tue Sep 29 16:00:52 UTC 2026
Updated every 50 seconds
Nagios® Core™ 4.4.6 - www.nagios.org
Logged in as ?

View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service

Service

HTTP

On Host

Terraform Nagios Target 01 (target1)

Member of

No servicegroups.

10.20.1.20

Service State Information

Current Status: **OK** (for 0d 1h 13m 47s)
Status Information: HTTP OK: HTTP/1.1 200 OK - 345 bytes in 0.001 second response time
Performance Data: time=0.000877s;0.000000;10.000000 size=345B;...0
Current Attempt: 1/3 (HARD state)
Last Check Time: 09-29-2026 15:57:05
Check Type: ACTIVE
Check Latency / Duration: 0.001 / 0.004 seconds
Next Scheduled Check: 09-29-2026 16:07:05
Last State Change: 09-29-2026 14:47:05
Last Notification: N/A (notification 0)
Is This Service Flapping? **NO** (0.00% state change)
In Scheduled Downtime? **NO**
Last Update: 09-29-2026 16:00:49 (0d 0h 0m 3s ago)

Active Checks: **ENABLED**
Passive Checks: **ENABLED**
Obsessing: **ENABLED**
Notifications: **ENABLED**
Event Handler: **ENABLED**
Flap Detection: **ENABLED**

Service Commands

✗ Disable active checks of this service

🕒 Re-schedule the next check of this service

❓ Submit passive check result for this service

✗ Stop accepting passive checks for this service

✗ Stop obsessing over this service

✗ Disable notifications for this service

📧 Send custom service notification

🕒 Schedule downtime for this service

✗ Disable event handler for this service

✗ Disable flap detection for this service

✗ Clear flapping state for this service

Service Comments

🗨 Add a new comment 🗑 Delete all comments

Entry Time	Author	Comment	Comment ID	Persistent	Type	Expires	Actions
This service has no comments associated with it							

Page Four

13.233.216.192/nagios4/

Machine LearningNeural Networks ar...Density-based clust...Fisher LDA-StanfordCompetitive Progra...CVB in Unsupervise...Red Hat AcademySystem Design Primer

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System

CommentsDowntimeProcess InfoPerformance InfoScheduling QueueConfiguration

Service Information

Last Updated: Tue Sep 29 16:01:08 UTC 2026
Updated every 50 seconds
Nagios® Core™ 4.4.6 - www.nagios.org
Logged in as ?

View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service

Service

NRPE Disk

On Host

Terraform Nagios Target 01 (target1)

Member of

No servicegroups.

10.20.1.20

Service State Information

Current Status: **OK** (for 0d 0h 38m 56s)
Status Information: DISK OK - free space / 5591 MB (72% inode=92%):
Performance Data: /-2140MB;6204.6979;0.7755
Current Attempt: 1/3 (HARD state)
Last Check Time: 09-29-2026 15:52:12
Check Type: ACTIVE
Check Latency / Duration: 0.000 / 0.065 seconds
Next Scheduled Check: 09-29-2026 16:02:12
Last State Change: 09-29-2026 15:22:12
Last Notification: 09-29-2026 15:22:12 (notification 0)
Is This Service Flapping? **NO** (11.32% state change)
In Scheduled Downtime? **NO**
Last Update: 09-29-2026 16:00:59 (0d 0h 0m 9s ago)

Active Checks: **ENABLED**
Passive Checks: **ENABLED**
Obsessing: **ENABLED**
Notifications: **ENABLED**
Event Handler: **ENABLED**
Flap Detection: **ENABLED**

Service Commands

✗ Disable active checks of this service

🕒 Re-schedule the next check of this service

❓ Submit passive check result for this service

✗ Stop accepting passive checks for this service

✗ Stop obsessing over this service

✗ Disable notifications for this service

📧 Send custom service notification

🕒 Schedule downtime for this service

✗ Disable event handler for this service

✗ Disable flap detection for this service

✗ Clear flapping state for this service

Service Comments

🗨 Add a new comment 🗑 Delete all comments

Entry Time	Author	Comment	Comment ID	Persistent	Type	Expires	Actions
This service has no comments associated with it							

Page Four

- Comments
- Downtime
- Process Info
- Performance Info
- Scheduling Queue
- Configuration

```

Current Status: OK (for 0d 1h 10m 5s)
Status Information: OK - load average: 0.00, 0.00, 0.00
Performance Data: load1=0 000; 0 150; 0 300; load5=0 000; 0 100; 0 250; load15=0 000; 0 050; 0 200; 1/3 (HARD state)
Current Attempt: 09-29-2026 15:53 19
Last Check Time: ACTIVE
Check Type: ACTIVE
Check Latency / Duration: 0 000 / 0 066 seconds
Next Scheduled Check: 09-29-2026 16:03 19
Last State Change: 09-29-2026 14:51 19
Last Notification: N/A (notification 0)
Is This Service Flapping? NO (0.00% state change)
In Scheduled Downtime? NO
Last Update: 09-29-2026 16:01 19 ( 0d 0h 0m 5s ago)

Active Checks: ENABLED
Passive Checks: ENABLED
Obsessing: ENABLED
Notifications: ENABLED
Event Handler: ENABLED
Flap Detection: ENABLED

```

- Disable active checks of this service
- Re-schedule the next check of this service
- Submit passive check result for this service
- Stop accepting passive checks for this service
- Stop obsessing over this service
- Disable notifications for this service
- Send custom service notification
- Schedule downtime for this service
- Disable event handler for this service
- Disable flap detection for this service
- Clear flapping state for this service

○ Add a new comment Delete all comments

Entry Time	Author	Comment	Comment ID	Persistent	Type	Expires	Actions
This service has no comments associated with it							